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Roll No

CE-803-GS

B.E. VIII Semester

Examination, December 2020

Grading System (GS)

Advanced Structure Design-II (Steel)

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions .
ii) All questions carry equal marks.

1. Design the central section for a welded plate girder having an effective span of 12m. It has to support a live load of 62kN/m and total dead load of 42kN/m. Take impact factor = $\frac{20}{14 + L}$, Where L is span in meters. Sketch the details.
2. a) Design the chimney shell for a self supporting steel stack of height 70m above foundation for the following data:
Diameter of cylindrical part = 3m, diameter at the base of flared part = 5.0m, height of flared part = 20m.
Take wind pressure = 0.90 kN/m² uniform throughout the height of stack.
b) Design the breech opening and base plate for above stack.
3. Design the container of 200kL suspended bottom overhead steel tank. Assume suitable shape for the tank.
4. Explain design component at trussed girder bridges for railways.

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5. Design a Steel Chimney (Self supported) of height 45m above the foundation and having its diameter at top is 3.5m thickness of brick lining is 115mm and this is supported by the chimney shell through the height. Chimney is situated in Bhopal city. The topography of the site is flat and the location of the terrain category 2. Compute wind loads on different zones of Chimney and design plates of the lowest zone.
6.
 - a) Design a circular elevated water tank for a capacity of 250 KL. The tank is supported over eight columns. Design conical roof, cylindrical portion and bottom hemispherical bottom plates.
 - b) Design a rectangular tank of capacity 110 kL of water. Plates of 1.25 m width and 8.5 m length are available. Design bottom plates and Tee covers.
7. A steel bunker 10m length and 8m width is required to store coal of bulk density 8000N/m^3 . The angle of internal friction is 35° . Height of vertical portion and hopper is 4m each. Design the side walls and through portions.
8. Write short notes on any two of the following:
 - a) Towers
 - b) Silo
 - c) Self supporting steel stacks
